NOTES

There aren't gameplay() and move() functions.

Game:

The implemented game is a board game that involves distributing stones across small cups on a board, with the goal of accumulating the most stones in the player's designated large cup by the end of the game. The game utilizes a 2D array to represent the board state and employs functions for initializing the game board and printing the board for visual representation during gameplay.

Part 1: initializeGame()

The `initializeGame()` function initializes the game board by placing a specified number of stones (`n`) into each of the small cups while leaving the large cups empty. The function uses a nested loop to iterate over the 2D array representing the board, setting the initial stone count in each small cup (`board[i][j] = n`) and ensuring the large cups (`board[i][6]`) start with zero stones.

```
void initializeGame(int board[2][7], int n) {
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < CUPS; j++) {
            board[i][j] = n;  // Set n stones in each cup
        }
        board[i][6] = 0;  // Large cups empty starts
    }
}
```

Part 2: printBoard()

The `printBoard()` function displays the current state of the board by printing the contents of each cup, including the stone count in each small cup and the number of stones in the designated large cups for both the player and the computer. The function uses formatted `printf` statements to visually represent the board layout.

```
void printBoard(int board[2][7]) {
    printf("+----+---+---+---+---+---+---+----+\n");
    printf("|    | %d | %d | %d | %d | %d | %d |    |\n", board[0][5], board[0][4], board[0][3], board[0][2],
board[0][1], board[0][0]);
    printf("|  %d |---+---+---+---+---+---| %d  |\n", board[1][6], board[0][6]);
    printf("|    | %d | %d | %d | %d | %d | %d |    |\n", board[1][0], board[1][1], board[1][2], board[1][3],
board[1][4], board[1][5]);
    printf("+----+---+---+---+---+---+---+----+\n");
}
```

Main Function:

The main() function demonstrates the usage of the initializeGame() and printBoard() functions to set up and display the game board. It also calculates and displays the final scores of both the player and the computer to determine the winner based on the number of stones in their respective large cups.

```
int main() {
    int board[2][CUPS + 1];
    int n = 4; // Number of stone to initialize each cups

    initializeGame(board, n);
    printBoard(board);


     // Determine the winner and print the result
    int p1score = board[0][CUPS];
    int pcscore = board[1][CUPS];

    if (p1score > pcscore) {
        printf("Player 1 wins, score is %d\n", p1score);
    }
    else if (pcscore > p1score) {
        printf("Computer wins, score is %d\n", pcscore);
    }
    else {
        printf("It is a tie\n");
    }

    return 0;
}
```

Output:

```
+----+---+---+---+---+---+---+----+
|    | 4 | 4 | 4 | 4 | 4 | 4 |    |
|  0 |---+---+---+---+---+---| 0  |
|    | 4 | 4 | 4 | 4 | 4 | 4 |    |
+----+---+---+---+---+---+---+----+
```

Conclusion:

As a result, the implemented game partially successfully initiates a board game situation, allowing the visual of the game board and determining the winner based on the final tile counts in the large cups. gameplay(); and move(); functions are missing. It may include its implementation as described in the assignment.